Established shifts in design practice

Mead–Conway VLSI

Chip + layout intent
→ scalable VLSI layout

Design rules + tools
+ fabrication access

RISC

Workload + software needs
→ hardware–software contract

Architectural principles
+ quantitative tests

RTL logic synthesis

Register-transfer behavior
→ gate-level implementation

Timing analysis
+ equivalence checking

Reusable generators

Parameters + design intent
→ reusable design family

Explicit constraints
+ implementation checks

Each abstraction gained value through repeatable ways
to produce implementations and check them.

Proposed AI-native design system

Research ambition

System intent
+ current
design inputs

→

Learned + conventional
methods, tools,
and checks

→

Evaluated alternatives
power, performance,
area, and limits